

Tribhuvan University
Institute of Science and Technology
2071

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 111)
(Digital Logic)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Long Questions:

Attempt any two questions: (2 × 10=20)

1. What are the various types of numbering system use in the digital logic? Explain.
Convert the $3EC8_{16}$ into different numbering system that you know.
2. Design the mod-6 asynchronous counter and explain with truth table.
3. What is demultiplexer? Draw its block diagram and explain its working principle.

Short Questions:

Attempt any eight questions: (8 × 5=40)

4. Convert the hexadecimal number 2BFC to binary and then to octal.
5. Proof the De-Morgan 1st and 2nd theorem with truth table and logic gates.
6. Simplify, the following Boolean function using three variable K-map.

a) $F(X,Y,Z) = \sum(0,3,2,5)$

b) $F(A,B,C) = \sum(0,2,4,5,6)$

7. Simplify the Boolean expression.

$$Y = \overline{A.B} + \overline{A} + \overline{B}$$

prepare truth table to show that the simplified expression is correct or not?

8. Explain the PLA (Programmable Logic Array).
9. How JK flip flop can convert into a D-flip flop?
10. What do you mean by synchronous counter? Explain with truth table.
11. Draw a 3 to 8 decoder circuit and explain its operation.
12. Mention the difference types of shift register and explain.
13. Write short notes on:-
 - a) CMOS
 - b) Universal gates
 - c) Error detection code